

Tutorial 6 – Advanced Function II

Passing of arrays

1. Create a function **toUppercase** that takes a character and returns its uppercase equivalent. Use this function to capitalize all characters in a character array representing a word.

The **main** function will prompt user for an input of string value, followed by conversion of all its letters to uppercase and display the result.

2. a) Develop a C++ function **inputSales** for user to input the monthly sales amount achieved over a year. The function will process the **sales** array for this purpose.

b) Write another C++ function **calculateAvgSales** of double type to process the **sales** array so that it will return the **average sales** obtained from the monthly sales of a year.

c) Prepare the **main** function where a **sales** array (double type) contains the sales figures of all the months of a year is declared. It will perform the input of the sales amount and finding the average monthly sales by calling the relevant functions above. Finally, it will display the average sales.
3. a) Write a C++ function **checkEven** which takes an integer number and return a boolean result if the number is even or not.

b) Do a **main** function which declares and initializes a vector array called **numbers** with a mixture of integer values. It will perform an even numbers checking using the above function and return the result in a table like below: -

Numbers	Even
-----	-----
13	No
18	Yes
24	Yes
35	No
:	:

Passing of structure

4. a) You are working on a hospital management system. A Patient structure contains various information like name, age, and body temperature. Write a function **checkFever** that takes only the temperature from the patient and determines whether the patient has a fever (above 37.5°C). The message "Fever detected!" or "Temperature is normal" will be displayed accordingly.

b) The main function will test the **checkFever** function a sample patient's temperature to be entered by user into a patient structure variable.

5. A courier service stores each package's base weight and adds packaging material based on the destination.
- Develop a function **computeTotalWeight** that uses a structure **Package** (with members **baseWeight** and **totalWeight**) and computes the totalWeight inside a function by adding the packagingWeight (passed by value) to baseWeight (in the structure, passed by reference).
 - Write a **main** function to prompt users for the base weight and packaging weight of a package to be delivered by the courier service. It should display the total weight required to deliver the package.

Storage classes

6. Study the following program:

1 #include <stdio.h>	14
2	15 void fun1(int y){
3 void fun1(int);	16 static int w = 2 ;
4 int fun2(void);	17 w *= y ;
5	18 cout << w;
6 int x = 3;	19 }
7	20
8 void main(){	21 int fun2(){
9 int limit = 1;	22 int z = 1 ;
10 for (; limit < 5 ; limit++)	23 return z += x ;
11 fun1 (limit);	24 }
12 cout << fun2();	
13 }	

- a) State the visibility and storage classes of x and z at the following points of program execution:

Variable	Visibility	Storage Class
x at line 6		
z at line 22		

- b) What is the output of the program?

7. Identify the storage classes and the scope for each of the variables in the following program skeleton and complete the table given below.

1 #include<stdio.h>	14 void f1(void)
2	15 {
3 int x;	16 int x;
4	17 ...
5 void main()	18 }
6 {	19
7 char c;	20 char d;
8 extern char d;	21
9 ...	22 float f2(int w)
10 }	23 {
11	24 static int v;
12 static float y;	25 ...
13	26 }

Variable Name	Line Number	Storage Class	Score (Line Number)
x	3		
c	7		
y	12		
x	16		
d	20		
w	22		
v	24		

8. Identify the storage classes and the scope of the following variables used in the program skeleton above.

<pre> 1 #include <stdio.h> 2 int K; 3 int main(void) 4 { int Q; 5 6 return 0 ; 7 }</pre>	<pre> 8 int funct1(int X, int Y) 9 { 10 int G; 11 static double C; 12 extern double B; 13 14 }</pre>	<pre> 15 double B; 16 void funct2(int A, int B) 17 { 18 int Z; 19 ... 20 }</pre>
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a) K (line 2)

b) G (line 10)

c) C (line 11)

d) B (line 12)

e) A (line 16)

Variable	Storage Class	Scope
K		
G		
C		
B		
A		

One variable in the above program is partly invisible within its own scope.

Give the **(i)** name of the variable and the **(ii)** line number where it is declared.

9. Write the output generated by the program below.

<pre> #include <stdio.h> int funct1 (int c); int funct2 (int c); int a = 10; b = 20; void main(void) { int count; for (count=1; count < 5; count++) cout << funct1(count); }</pre>	<pre> int funct1 (int x) { int c, d; c = funct2(x); d = (c > 10) ? b : (a + c); return (d); } int funct2 (int x) { static int pr = 1; pr = pr * x; return (pr); }</pre>
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